

10 kW Orbital Edge Node

Public Mission Definition and Pre-SRR Baseline

Compute where space data begins.



Notional concept render. This is not flight CAD and does not establish interfaces, dimensions or launch compatibility.

A first owned LEO system sized around 10 kW continuous compute-payload input. This package defines the mission and verification intent; it is not flight CAD, an ICD or a launch-provider compatibility statement.

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PRE-SRR BASELINE

1. Mission statement

TOP-LEVEL

REQUIREMENT

The mission shall demonstrate customer-relevant processing of space-originated data while delivering 10 kW continuous electrical power at the compute-payload input during nominal operations, including the design eclipse where applicable.

10 kW

PAYLOAD INPUT

500-600 km

LEO TRADE SPACE

3-7 t

WET-MASS ASSUMPTION

Parameter	Public baseline	Status
Mission class	First owned orbital edge-processing node	REQUIREMENT
Orbit	550 km circular LEO screening case; SSO local time TBD	WORKING ASSUMPTION
Nominal mission life	5 years; 7-year design sensitivity	WORKING ASSUMPTION
Compute payload	10.0 kW continuous at DC input	REQUIREMENT
Full spacecraft load	12-15 kW; nominal 13.5 kW	WORKING ASSUMPTION
Wet launch mass	3-7 tonnes; nominal planning point 4.7 tonnes	WORKING ASSUMPTION
Launch class	Heavy rideshare or dedicated	TBD BY SUPPLIER
First market	Defense/sovereign, EO, maritime and disaster-response outcomes	BUSINESS HYPOTHESIS

WHAT THE FLIGHT MUST PROVE

2. Mission objectives

ID	Objective	Success evidence
MO-01	Commission and maintain positive spacecraft control	Healthy power, attitude, communications and thermal telemetry
MO-02	Operate the compute system at 1, 3-5 and 10 kW levels	Stable staged ramp with no limit violations
MO-03	Sustain 10 kW continuous payload input	Agreed-duration run through the design environmental case
MO-04	Process a customer-defined space-originated dataset	Accepted quality, latency and data-reduction results
MO-05	Demonstrate fault containment and recovery	Injected hardware, software and link failures recover within limits
MO-06	Demonstrate secure workload and data lifecycle	Signed load, provenance, encryption, access and deletion evidence
MO-07	Preserve end-of-life disposal and passivation	Verified maneuver, energy and command reserve

Non-objectives

- Proving commodity cloud economics or serving arbitrary public workloads.
- Operating in GEO or at a Lagrange point.
- Manufacturing every sensor, spacecraft subsystem or launch vehicle.
- Claiming 10 kW of accelerator-silicon consumption.
- Scaling to 100 kW before first-node customer and operations evidence exists.

SENSOR TO DECISION

3. Concept of operations

Phase	Activities	Critical constraint
Ground preparation	Customer workload, data rights, model, keys and acceptance plan are frozen	No uncontrolled model or data reaches flight
Launch and early orbit	Deploy, establish control, inspect power and thermal state	Compute remains disabled until platform health is proven
Compute commissioning	Ramp one tile, then multiple tiles, then full cluster	Each step has automatic abort limits
Service operation	Receive sensor data, process, store, downlink result or cross-task	Data path and customer priority are scheduled
Contingency	Throttle, isolate tile, reboot domain, enter safe mode	TT&C and disposal capability are protected
End of life	Deorbit, passivate and close data custody	Regulatory and customer obligations remain enforceable

Data-path alternatives

Architecture	Advantage	Primary risk
Same-platform sensor	Simpler ingress and timing	Sensor integration can dominate spacecraft configuration
Optical crosslink from partner spacecraft	Shared node can serve multiple operators	Acquisition geometry, terminal interoperability and cost
High-rate RF crosslink	Potentially broader partner compatibility	Spectrum, antenna size, interference and licensing
Store-and-forward upload	Useful for software and limited test data	Does not establish the core space-originated-data advantage

LEVEL-1 AND LEVEL-2 INTENT

4. System requirements baseline

ID	Requirement	Verification
SYS-001	The system shall deliver 10.0 kW continuous at the compute-payload DC interface in nominal operation.	Test + analysis
SYS-002	The system shall protect TT&C, attitude control and disposal from compute-domain failures.	Test + analysis
SYS-003	The system shall measure power, temperature, flow and error state at tile and domain level.	Inspection + test
SYS-004	The system shall isolate and restart a failed compute tile without full-node loss.	Fault-injection test
SYS-005	The system shall maintain a cryptographically controlled workload and data chain.	Inspection + demonstration
SYS-006	The system shall process at least one customer-defined workload to agreed acceptance metrics.	Demonstration
SYS-007	The system shall support a safe throttled mode when power, thermal or link constraints are exceeded.	Test
SYS-008	The system shall retain deorbit and passivation capability after credible single failures.	Analysis + test
SYS-009	The system shall expose versioned electrical, mechanical, thermal, data and operational interfaces.	Inspection

An Interface Control Document, or ICD, will define the agreed boundary between two parties or subsystems: voltages, connectors, data protocols, thermal rejection, geometry, loads, responsibilities and verification. This mission definition precedes the ICD; it does not substitute for one.

BETA-ZERO 550 KM SCREEN

5. Rev C sizing envelope

Metric	Planning range	Nominal	Evidence class
Full spacecraft load	12-15 kW	13.5 kW	WORKING ASSUMPTION
Solar-array BOL	29.87-37.34 kW	33.61 kW	CALCULATED
Active-PV equivalent	99.57-124.46 m2	112.02 m2	CALCULATED
Gross solar planform	124.46-155.58 m2	140.02 m2	CALCULATED
Battery nameplate	25.16-31.45 kWh	28.31 kWh	CALCULATED
Heat rejection	11.64-14.55 kW	13.10 kW	CALCULATED
Equivalent radiator area	33.26-41.57 m2	37.41 m2	CALCULATED
Physical radiator planform	35-60 m2	TBD	WORKING ASSUMPTION
Wet launch mass	3-7 tonnes	4.7 tonnes	WORKING ASSUMPTION
Approximate deployed span	25-40 m	TBD	WORKING ASSUMPTION

The calculated screen is not a subsystem allocation. It omits detailed incidence, temperature, harness, converter, structural, deployment, contamination, attitude, shielding and propellant effects. Supplier-backed budgets must replace the planning ranges by PDR.

SUPPLIER

BOUNDARY

Current public bus claims are leads, not closure. Apex publicly advertises 8 kW orbit-average payload power and 6 kW continuous heat dissipation for Comet Mini. EnduroSat advertises scalable generation and a multi-tonne payload class, but its public values do not prove a turnkey 10 kW continuous compute service.

PRELIMINARY OWNERSHIP

6. Subsystem allocation and interfaces

Subsystem	Preliminary responsibility	Interface to freeze by PDR
Compute payload	Satellite Inference	DC power, heat pickup, data, timing, telemetry and mechanical envelope
Bus and primary structure	Bus supplier / co-development	Mass properties, stiffness, loads, volume and safe-mode services
Solar and PMAD	Array and EPS suppliers	BOL/EOL power, voltage, MPPT, pointing, deployment and protection
Battery	EPS/battery supplier	Energy, voltage, cycle life, DoD, thermal and fault containment
Thermal transport and radiator	Co-developed with thermal supplier	Heat load, temperatures, flow, redundancy, view factor and deployment
ADCS and propulsion	Bus supplier	Flexible-body modes, momentum, pointing, maneuver and disposal
Communications	Bus/comms suppliers plus Satellite Inference	RF/optical links, data rates, acquisition, encryption and scheduling
Ground and mission operations	Satellite Inference plus service providers	Command authority, telemetry, security, customer API and incident response
Launch integration	Launch integrator and provider	Envelope, loads, coupled loads, separation and range documentation

WHAT CAN BE PROVEN BEFORE FLIGHT

7. Ground development and verification matrix

Test article	Test	Pass criterion	Flight risk reduced
1 kW tile	1,000-hour representative workload	No unplanned reset; stable power, temperature and throughput	Compute and local cooling
1 kW tile	Fault injection	Errors contained; watchdog and restart meet limit	Radiation-like upset response
1 kW tile	Power transient	No unsafe bus interaction or data corruption	EPS interface
10 kW breadboard	Full-load steady state	10 kW payload input with closed heat balance	System thermal transport
10 kW breadboard	N-1 tile and pump case	Graceful degradation and protected platform loads	Fault tolerance
Network rig	Sensor-to-result flow	Customer p95 latency, accuracy and data reduction met	Commercial mission closure
Engineering model	EMI/EMC and interfaces	Margins against allocated limits	Integration
Structural article	Modal, vibration and deployment	Correlated model and positive margins	Launch and deployment
Thermal-vacuum article	Hot/cold and transitions	No limit violation; correlated thermal model	Orbital environment
Flight system	End-to-end mission rehearsal	Nominal and contingency procedures accepted	Operations

Software acceptance metrics

- p95 sensor-to-decision latency under defined data volume and link conditions.
- Recall at the customer's maximum acceptable false-alarm rate.
- Accepted outputs per kWh and raw-to-result data-reduction ratio.
- Availability by workload class and recovery time by failure class.
- Cryptographic identity, provenance, deletion and audit evidence.

NO LAUNCH BOOKING BEFORE GEOMETRY

8. Reviews, schedule logic and decision gates

Gate	Minimum package	Go decision
MDR	Mission need, workload and data path, LEO trade, first customer hypothesis	Mission is coherent enough for requirements
SRR	Requirements, tile test, initial suppliers, compliance plan and V&V map	Requirements are necessary and testable
PDR	10 kW breadboard, selected architecture, budgets, CAD concept, ROMs and customer evidence	Architecture can meet requirements within margin
CDR	Released design, analyses, qualification procedures, ICDs and launch interface	Design is ready to build and qualify
FRR	Qualified hardware, licenses, operations, range and launch readiness	Risk is acceptable for launch

Launch price cannot be meaningfully requested as a binding quote until mass, stowed envelope, separation system, orbit, schedule, hazardous materials and integration responsibilities are known. Early launch-provider discussions are still useful for planning envelopes, compatible dispensers, lead time and documentation expectations.

MDR CLOSURE LIST

9. Open items and required decisions

Decision	Options	Required evidence
Same-platform sensor or crosslinked node	One sensor, multiple sensors, hybrid	Customer and partner data-path study
Ordinary SSO or high-beta trade	Mid-morning/afternoon, dawn-dusk sensitivity	Lighting, beta, thermal and launch availability
Single spacecraft or distributed pair	One 10 kW node, two 5 kW nodes	Mass, reliability, network and cost trade
Compute hardware	Accelerator, CPU, memory, storage and redundancy mix	Customer benchmarks plus radiation screen
Thermal architecture	Pumped loop, heat pipes, cold plates, deployable radiators	Thermal model and supplier study
Launch class	Heavy rideshare or dedicated	Supplier-backed mass properties and envelope
Operational customer	Defense/sovereign, EO partner, maritime or disaster program	Paid evaluation and conditional reservation

Sources

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